

# Pranav Karawale

Precision *beyond* pixels.

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## SUMMARY

Founding Engineer and CS student building edge-deployed IoT pipelines and applied ML systems for industrial clients. Proven track record of scaling consumer applications and optimizing cloud infrastructure.

## SKILLS

**Languages:** Python, TypeScript, Go, C, Bash      **Infra & DevOps:** AWS (EC2, S3), Docker, Traefik, Grafana, Prometheus  
**Backend:** Node.js, FastAPI, PostgreSQL, MongoDB, MQTT, GRPC      **AI & Edge:** YOLO, OpenCV, Raspberry Pi, STM32, Modbus/PLC, PlatformIO

## EXPERIENCE

### Pravideon Pvt Ltd

*Founding Engineer*

**Jan. 2024 – Present**

Navi Mumbai, India

- Worked on designing AI models and controls for industrial fire monitors to detect and respond to fire hazards autonomously. Collaborated with stakeholders to build a secure, real-time remote monitoring dashboard.
- Built live data pipeline using Python and PostgreSQL, indexing 15 million+ historical records in under 24 hours, and ingesting daily new data.
- Developed real-time IoT monitoring dashboard deployed across 10+ industrial sites, achieving 99.9% uptime.
- Created AWS-hosted ingestion pipeline processing and indexing 200K+ rich-media questions for enterprise LMS.

### Avignon Université

*Research Intern*

**Jun. 2025 – Aug. 2025**

Avignon, France

Designed new authenticated encryption with associated data encapsulation scheme for LoRaWAN for protection against various MiTM, replay attacks, and developed vector database based reinforcement learning for auditing end devices.

### Freelancing

*AI Tooling Engineer*

**Apr. 2023 – Dec. 2023**

Remote

Architected a LLM agnostic prompt orchestration framework utilizing React JSX as a declarative language to build, manage, and scale complex agentic workflows across different LLM providers (OpenAI, Anthropic, Gemini).

## EDUCATION

### Vishwaniketan's iMEET (University of Mumbai)

*B.E. Computer Engineering*

**Nov. 2022 – Jun. 2026**

## PROJECTS

### Newage Fire AI

Autonomous Fire detection and response system. Uses Modbus and custom trained models to target and extinguish fires using industrial monitors, and send live alerts.

**Ixora** <http://github.com/retronav/ixora>

An extension pack to make interactive WYSIWYG markdown editors using CodeMirror 6.

### DuckThisFile

Self-hosted file portal with microservice architecture for image and PDF processing with privacy.

## ACHIEVEMENTS

**Winner:** Echelon '25 Hackathon (AI P&ID Digitization)

**Gold Medallist:** JuniorSkills India 2021 (Web Technologies - National Level)

**Leadership:** Mentored peers & led technical workshops as Technical Lead, GDG on Campus ViMEET (2024-25).

**OSS Contributor:** Astro [<https://astro.build>]